

CREDIT CARD TRANSACTION ANALYSIS

Project Overview:

- Explored 100K+ credit card transactions using SQL to uncover customer spending patterns and business insights.
- Applied advanced SQL techniques — JOINS, CTEs, window functions, and aggregations.
- Analyzed transaction behavior in depth across multiple dimensions such as customer demographics, card types, cities, and expense categories to identify key drivers of spending variation.

Q1: Top 5 cities with the highest spends and their percentage contribution to total credit card spends.

CITY	EXPENSE	TOTAL SPENT	PERCENTAGE CONTRIBUTION
Greater Mumbai	576,751,476	4,074,833,373	14.15%
Bengaluru	572,326,739	4,074,833,373	14.05%
Ahmedabad	567,794,310	4,074,833,373	13.93%
Delhi	556,929,212	4,074,833,373	13.67%
Kolkata	115,466,943	4,074,833,373	2.83%

Q2: Highest spend month and amount spent in that month for each card type.

CARD TYPE	YEAR	MONTH	MONTHLY EXPENSE	RANK
Gold	2015	January	55,455,064	1
Platinum	2014	August	57,936,507	1
Signature	2013	December	58,799,522	1

Silver	2015	March	59,723,549	1
--------	------	-------	------------	---

Q3: Transaction details for each card type when it reaches a cumulative total of 1,000,000 spends.

TRANSACTION ID	CITY	TRANSACTION DATE	CARD TYPE	EXPENSE TYPE	GENDER	AMOUNT	CUM SUM
1522	Delhi	04-10-2013	Gold	Food	M	281,924	1,272,624
191	Ahmedabad	05-10-2013	Platinum	Bills	F	1,537,482	612,572
73	Delhi	04-10-2013	Signature	Bills	F	550,782	1,285,819
7565	Bengaluru	04-10-2013	Silver	Food	F	205,179	1,115,582

Q4: Top 10 cities that had the lowest percentage spend for the gold card type.

CITY	TOTAL SPEND	GOLD SPEND	GOLD CONTRIBUTION (%)
Dhamtari	425,241	1,416	0.33%
Solan	790,741	2,910	0.36%
Suar	861,153	3,328	0.38%
Bhiwandi	557,886	2,186	0.39%
Bhadrachalam	1,133,551	4,544	0.40%
Lalitpur	848,581	3,482	0.41%
Uravakonda	772,163	4,131	0.53%
Ranchi	619,502	3,547	0.57%
Brahmapur	539,133	3,184	0.59%

Samastipur	741,869	5,059	0.68%
------------	---------	-------	-------

Q5: City, Highest Expense Type, and Lowest Expense Type.

CITY	HIGHEST EXPENSE TYPE	LOWEST EXPENSE TYPE
Achalpur	Grocery	Entertainment
Adilabad	Bills	Food
Adityapur	Food	Grocery
Adoni	Bills	Entertainment
Adoor	Fuel	Bills
Afzalpur	Fuel	Food
Agartala	Grocery	Food
Agra	Bills	Grocery
Ahmedabad	Bills	Grocery
Ahmednagar	Fuel	Grocery

* Showing sample of top results (976 more rows in full dataset)

Q6: Percentage contribution of spends by females for each expense type.

EXP TYPE	TOTAL SPEND	FEMALE SPEND	FEMALE CONTRIBUTION (%)
Entertainment	726,437,536	358,663,333	49.37%
Fuel	789,135,821	392,282,421	49.71%
Grocery	718,207,923	365,646,998	50.91%

Travel	109,255,611	55,865,530	51.13%
Food	824,724,009	452,817,279	54.91%
Bills	907,072,473	580,035,469	63.95%

Q7: Card and expense type combination with the highest month-over-month growth in Jan-2014.

CARD TYPE	EXP TYPE	YEAR	MONTH	TOTAL SPEND	PREV MONTH SPEND	MOM GROWTH
Platinum	Grocery	2014	January	11,225,634	7,757,562	4,498,781

Q8: City with the highest total spend-to-total number of transactions ratio during weekends.

CITY	RATIO
Sonepur	299,905
Lanka	298,960
Shirpur-Warwade	298,572
Zunheboto	298,567
Puranpur	298,518
Yawal	298,469
Margao	296,975
Amroha	296,502
Jatani	294,951
Modinagar	294,699

* Showing sample of top results (797 more rows in full dataset)

Q9: City that took the least number of days to reach its 500th transaction.

CITY	FIRST TRANSACTION	LAST TRANSACTION	DAYS TO 500
Bengaluru	04-10-2013	24-12-2013	81
Greater Mumbai	04-10-2013	28-12-2013	85
Ahmedabad	04-10-2013	29-12-2013	86
Delhi	04-10-2013	02-01-2014	90
Chennai	04-10-2013	19-10-2014	380
Hyderabad	04-10-2013	23-10-2014	384
Kolkata	04-10-2013	28-10-2014	389
Lucknow	05-10-2013	30-10-2014	390
Kanpur	04-10-2013	03-11-2014	395
Jaipur	04-10-2013	09-11-2014	401
Surat	04-10-2013	11-11-2014	403
Pune	04-10-2013	19-11-2014	411